Report

A pre-registered *in silico* falsification programme on a redox-switch model of parvalbumin-interneuron damage

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Abstract. We built a mechanistic ODE model of oxidative damage to parvalbumin (PV) interneurons — a bistable peroxiredoxin/sulfiredoxin redox switch, a two-stage inducible defence ($\tau_m = 12\,\mathrm{h}$, $\tau_s = 63\,\mathrm{h}$) with priming, asymmetry, and per-cell capacity, a Phillips–Robinson sleep–wake switch, and a Wilson–Cowan cortex closing the loop through disinhibition — and spent most of the effort trying to break it. Every hypothesis was written as a numbered rule; every threshold was derived from expectations and variances before the computation; verdicts sit in an append-only ledger and are never revised. Of 128 pre-registered parts across 26 rules, 66 passed, 48 failed, 13 were unfalsifiable, one fell outside its own analytic bracket. The model describes the generation and termination of an episode under oxidative load; the failures are the main result.

1. Object of study

Hypothesis. PV interneurons hold cortical excitation in check, set the gamma rhythm, and pay with the highest energy turnover in the network, hence a large peroxide flux. Part of the antioxidant defence is produced only while the NMDA receptor — the cell's activity sensor — works; a weak receptor means less defence and more oxidants, and oxidants suppress the receptor: a positive loop. The overloaded PV cell does not die but *silences* (less parvalbumin, less GAD67), the cortex disinhibits, load rises: a second loop; a disinhibited hippocampus raises dopamine responsivity (the Grace circuit, in which hippocampal output drives the dopamine system) — the bridge to positive symptoms. Each arrow is supported experimentally; the chain as a whole never has been. The quantitative question: is the loop gain large enough to *close*, i.e. to make the cell bistable? Bistability is what converts slow accumulation into a sharp transition with hysteresis — the clinical shape (long prodrome, abrupt first episode, episodes of hours–days).

Strategy. Build the minimal model; then falsify it systematically: numbered rules with parts, thresholds from analytic expectations only, append-only verdict ledger, bitwise-verified code, frozen reference implementation.

2. The model

The model is a stack of four layers with separated scales: the cell (minutes–days), the sleep switch (hours), the cortex (adiabatic), the population (via one scalar).

2.1. Redox core

Let $x \in [0,1]$ be the working fraction of peroxiredoxin, c the receptor-maintained protection, u the peroxide, ν the load. u is fast (algebraic balance); the dynamics are

$$\begin{aligned} &\nu = \varepsilon u + \frac{xu}{1+u}, \qquad \varepsilon = e_0 c^{\gamma}, \qquad N = \frac{N_0}{1+u/K_N}, \\ &\dot{x} = c^{\eta}(1-x) - \mu_0 \frac{xu^2}{1+u}, \qquad \dot{c} = \frac{1}{T}\Big(N + \frac{\beta u}{K_B+u} - c\Big). \end{aligned} \tag{1}$$

The first line and the x-equation are measured kinetics (Selvaggio et al. 2018); the c-equation is an assumption of form. The core is a two-variable reduction of the Selvaggio–Coelho–Salvador model [3]: the thioredoxin system is in excess (so it collapses into an effective repair rate), peroxide is adiabatically eliminated, and the slow sulfiredoxin induction becomes c. Concretely: the peroxide ODE of the full kinetics, $\dot{u} = \nu - \varepsilon u - xu/(1+u)$, is collapsed by time-scale separation (u relaxes in seconds) into the algebraic balance $\nu = \varepsilon u + xu/(1+u)$ — production equals clearance. This balance is a quadratic in u, so at each instant the load ν (the model's input, not a state variable) fixes u implicitly, and u is the only channel through which ν acts on x and c. In the simulation the per-cell load is $\nu = L(t)\,v_j$, with $L(t)$ the set-level load process of section 2.6 and v_j the cell multiplier of section 2.5. Notation:

- e_0: scale of the non-enzymatic clearance εu; μ_0: scale of the two-hit hyperoxidation $\mu_0 x u^2/(1+u)$ relative to repair;
- η: exponent of the defence-amplified repair $c^{\eta}(1-x)$; γ: exponent of the bypass clearance $\varepsilon = e_0 c^{\gamma}$;

- K_N: saturation of the peroxide suppression of the receptor-maintained defence $N = N_0/(1+u/K_N)$, with $N_0 < 1$ a congenitally weak receptor;
- β, K_B: strength and half-saturation of the stress induction $\beta u/(K_B+u)$;
- T: defence-protein lifetime in repair-time units.

The provenance of each term is documented in the repository.

Proposition 2.1 (Bistability)**.** *Over-oxidised peroxiredoxin is repaired slowly, so working enzyme* decreases *with peroxide while clearance* requires *enzyme: the feedback in* (1) *produces two saddle-node folds and a hysteresis window of loads in which "defence on" and "defence off" are both stable. Bistability of the fast layer (defence c frozen) is exact: with $\mu = \mu_0/c^\eta$ the nullcline $\nu(u) = \varepsilon u + u/(1+u+\mu u^2)$ is non-monotone iff*

$$\varepsilon < \Phi(\mu), \qquad \Phi(\mu) = \max_{u>0} \frac{\mu u^2 - 1}{(1+u+\mu u^2)^2}, \tag{2}$$

since $\frac{d\nu}{du} = \varepsilon + (1-\mu u^2)/(1+u+\mu u^2)^2$; the folds are the extrema of $\nu(u)$, and the slow variable c moves them without creating or removing the window. Measured cells (the 15 cell lines of Selvaggio et al. [3]) lie outside the window; entering it requires repair weakened 1.6–22*-fold — the proposed state of PV cells under weak NMDA signalling.*

Parameter sets. Nine parameters are sampled (e_0, μ_0, K_N, K_B, T log-uniform; $\eta \in [1,2]$, $\gamma \in [0,1]$, $N_0 \in [0.3,1]$ uniform) with $\beta = N_0 \cdot U(0,3)$ (measured induction up to 2.85×). A set is accepted iff bistable: window ratio > 1.05 and two stable states at the geometric-mean load. These accepted sets are the model's *potential patients*: only a set whose cell already sits in the bistable regime — the weakened-defence state the hypothesis postulates for PV cells under weak NMDA signalling — can develop the switch-mediated damage the model describes; everything else in the sampled space is healthy by construction and is not studied. $K = 200{,}000$ candidates $\to n_{\text{sel}} = 2400$ accepted sets (seeded, cached); the resting load sits at $q = 0.10$ of the window.

2.2. Inducible defence

Two stages. Induction signal m, then protein s, added to c:

$$\dot m = \frac{1}{\tau_m}\Big(\beta\frac{u}{K_B+u} - m\Big), \qquad \dot s = \frac{1}{\tau_s}(m-s), \qquad \tau_m = 12\,\text{h},\ \tau_s = 63\,\text{h}, \tag{3}$$

with a step response that is always sigmoidal; half-level at 57 h. The stationary state (hence window, folds, gap law) is unchanged; only the transient changes.

Priming. A trace p on the induction signal amplifies it: $\dot p = (u/(K_B+u) - p)/\tau_p$, $\beta_{\text{eff}} = \beta(1+Ap)$, $A = 2$ (ceiling ×3, measured; τ_p scanned — unmeasured).

Asymmetry. Rise and decay constants of s are split, $r = \tau_\downarrow/\tau_\uparrow$.

Proposition 2.2 (Asymmetric integrator)**.** *For a two-level daily input, high for a fraction f of the day, the stationary mean of the integrating variable stands at*

$$w(r) = \frac{fr}{fr+1-f}, \tag{4}$$

$f = 0.705$ (measured wake fraction). Asymmetry turns the integrator into a peak detector ($r > 1$) or trough detector ($r < 1$); no fitted numbers enter (4)*.*

Capacity loop. Per cell, while damaged, $N_0 \to N_0(1-a_n)$ (loss 1 d), recovery slow (30 d): a latch on recovery, not on load. *Dopamine loop*: a tone d driven by disinhibition $E/E_r - 1$ (excitation above its resting level E_r) or by the damaged fraction D ($\tau_d = 7$ d; D is defined in section 2.6) multiplies the load by $(1 + a_d d)$; at $a_d = 0$ the model is reproduced bitwise.

2.3. Sleep–wake switch

Phillips–Robinson, "new nominal" set (Skeldon et al. 2014), taken whole, untuned:

$$
\begin{aligned}
&Q(V) = \frac{Q_{\max}}{1 + e^{-(V-\theta)/\sigma_Q}}, \quad \tau_V \dot{V}_v = \nu_{vm} Q(V_m) + D_v - V_v, \quad \tau_M \dot{V}_m = A_m + \nu_{mv} Q(V_v) - V_m, \\
&D_v = B_0 + \nu_{vc} \cos \tfrac{2\pi(t-\alpha)}{24} + A_G G - k_d D_{\mathrm{cpl}}, \quad \dot{G} = \begin{cases} (1-G)/18.2\,\mathrm{h} & \text{awake}, \\ -G/4.2\,\mathrm{h} & \text{asleep}, \end{cases}
\end{aligned}
\tag{5}
$$

wakefulness := $Q(V_m) > 1\,\mathrm{s}^{-1}$. Here Q is the firing-rate sigmoid ($Q_{\max} = 100\,\mathrm{s}^{-1}$, threshold θ, slope σ_Q); V_v, V_m the sleep- and monoaminergic-nucleus voltages with time constants τ_V, τ_M and mutual inhibition ν_{vm}, ν_{mv}; D_v the drive to the sleep nucleus — constant part B_0, circadian amplitude ν_{vc} with phase α, somnogen weight A_G; G the somnogens; $-k_d D_{\mathrm{cpl}}$ the sleep–damage coupling, with D_{cpl} the mean damage or the damaged fraction. Checks against publication: folds at 1.450/2.463 mV; free cycle 15.5/8.5/24.0 h; somnogen swing 0.527 vs. 0.50 — unfitted. Damage enters via $-k_d D_{\mathrm{cpl}}$ ($k_d = 2\,\mathrm{mV}$): a damaged population suppresses the sleep nucleus and lengthens wakefulness.

2.4. Cortex

Wilson–Cowan pair (E excitatory, I PV), with gain a and threshold θ suppressed in the notation, $F(z) = 1/(1 + e^{-a(z-\theta)})$:

$$
\dot{E} = -E + F(c_1 E - c_2 g I + P_e), \qquad \dot{I} = -I + F(c_3 E - c_4 g I), \qquad g = 1 - 0.4\,D,
\tag{6}
$$

$P_e = P_{e0} + P_{e1} \min(Q(V_m)/Q_{\mathrm{wake}}, 1)$; here c_1–c_4 are the connectivity weights, P_e the wakefulness-borne drive to E, Q_{wake} its wake threshold. The multiplier on *all* PV output weights with $g_{\mathrm{dep}} = 0.4$ is from Volman et al. 2011 (inhibition to 60 %). Cortical scales are 10^3 times faster than everything else: stationary solution by damped iteration (adiabatic elimination); the gamma rhythm is absent by design. At full damage E rises 19 % — a consequence of two literature numbers, not a fitted gain.

2.5. Population

$J = 32$ cells differing in one scalar, the load multiplier $v_j = \exp(\sigma_v z_{p_j})/\mathrm{mean}$: $\sigma_v = 0.10$ is the spread and z_{p_j} the fixed normal quantiles of the edge-pinned probability grid (pinned to $J = 32$). Cells couple only through network activity (load on survivors rises with the damaged fraction). Gradualness is *from the spread, not the network* (F5, refuted). The grid subtlety: J is a model parameter through the realised spread, $\mathrm{std}(\ln v) = 0.1226/0.1060/0.0980/0.0943$ at $J = 8/16/32/64$ vs. 0.0910 in the limit ($O(1/J)$ convergence), with log-sensitivity ≈ 2.5 (section 4.4).

2.6. Load, episodes, and the gap law

The load $L(t)$: base at $q = 0.10$ of the window; circadian swing 20 % — the weight A_H of the somnogen deviation $G - \bar{G}$ in the load, calibrated on the switch; fast noise $\sigma = 0.08$ on log-load, correlation $0.3T$; slow noise (σ_s) *off* — withdrawn in F6, F18, but σ_s stays in the band budget; optional serotonin bursts ($\times 1.5$, rate 0.1/day, decay 2 h, ceiling $\times 2$); auxiliary protocols for the loop rules (two-phase break–release; static multiplier n_{mul} as the loop's static equivalent).

Definition 2.3 (Episode, band)**.** *An episode is the time with damaged fraction $D > 1/2$ (the continuous mean-damage variable for cross-J comparisons; crossing moment interpolated inside the step). The* band *is the set of parameter sets whose upward gap is covered by the available burst and whose downward gap fits the noise budget $2\sigma + \sigma_s$.*

Proposition 2.4 (Gap conservation)**.** *With window ratio W, load position q (log scale), feedback factor κ, daily peak and trough P_+, P_-:*

$$
\ln \frac{W^{1-q}}{P_+} + \ln(W^q \kappa P_-) = \ln \frac{W \kappa P_-}{P_+} \quad (q\text{-independent}).
\tag{7}
$$

Typical sets: sum $\approx 9\,\sigma$ of load noise; the clinical picture (rare episodes of hours–days) needs $\approx 4.5\,\sigma$. *The same noise cannot both break rarely and release quickly.* The exits: a one-sided trigger, or a variable defence that raises its own lower boundary — the pursued route (section 2.2).

3. Computation

The computation is organised around the rules: each hypothesis is one rule file, blocks are requested through shared constructors and hash-deduplicated (78 requested → 40 unique), noise is drawn per block so blocks are strictly paired, and every part records its verdict in an append-only ledger (date and block hashes) — verdicts are never revised; a defect is annotated beside the verdict and re-asked as a new rule on a fresh seed. Run commands and the file layout live in the repository README (linked in the title).

3.1. Statistics

A criterion is usable only if its null value, its noise at our sample size, and its power are known before the run:

$$\mathrm{MDE} = (z_{0.95} + z_{0.80})\,\mathrm{SE} = 2.4865\,\mathrm{SE}; \tag{8}$$

agreement claims use a 3 SE tolerance that must exceed the MDE, or the claim is unfalsifiable in principle. The sign test carries a *mandatory direction*; medians use the one-sided order-statistic bound; gaps use within-set CV^2 (null exactly one); every criterion self-tests on synthetic data with known answers before any run. The analytic risk model (one model for all rules): the log-load sits at $m_0 = \ln(\mathrm{lo}\cdot(\mathrm{hi}/\mathrm{lo})^q)$, the barrier at $\ln(\mathrm{hi})$, so the log-gap is $\Delta = (1-q)\ln(\mathrm{hi}/\mathrm{lo})$; an onset is a noise excursion over the barrier, rate $h(t) = e^{-d(t)^2/2}$ with $d(t) = (\Delta - \ell(t))/\sigma$, where $\ell(t)$ is the circadian rise of the log-load (piecewise linear, zero-mean over the day); the pre-exponential cancels in all ratios. This one model yields, unfitted: the late-wake onset share (F2a: 0.856 predicted, 0.885 measured), the swing-response curve, the mid-scale fork (F5), the burst-trigger share (F8a), the wake-lengthening ratio from the deterministic switch (F4a, F6a), the affine duration law (F7, F8b). A rank correlation across eight bins has error $1/\sqrt{7} = 0.38$; the fix is per-set slopes with attenuation $\rho_{\mathrm{set}} = \rho_{\mathrm{pool}}/\sqrt{1+\lambda}$ — in the real data $\lambda = 33.8$ turned an MDE of 0.94 into 0.076 and closed four of five parts.

3.2. Reproducibility

(i) *Frozen monolith*: the regression suite requires the set sampling, band arithmetic, and `simulate` to match the frozen monolith bitwise on pinned configurations; cross-machine comparison uses 10^{-12} with a machine fingerprint (measured worst 3×10^{-15}). (ii) *Fused kernel*: assembled from the reference source by textual substitution with uniqueness checks (an edit breaks the assembly, never diverges silently); verified on all 67 output fields over nine pinned configurations (counting bitwise; worst continuous deviation 9.5×10^{-15} vs. 10^{-12}) and by diffing all verdicts; 3.3× wall-clock gain. (iii) *Step size*: continuous means converge to 0.5 % at $M = 800$; counting quantities do not — the Richardson limit is 2.0 % above the value at $M = 1600$ — so counting thresholds finer than ≈ 3 % are not trusted, stated rather than hidden.

3.3. Ten statistical traps

Each once produced a false conclusion in our hands; each is closed by a code check: (1) median over sets of a rare-event share is degenerate; (2) the share of short gaps $\equiv 1 + \mathrm{CV}^2$ of set rates — physics-free; (3) sign-test direction is a mandatory argument; (4) zero error by construction in self-normalised columns; (5) a saturating measured quantity drowns real effects — measure the rate, not the ever-share; (6) an equivalence tolerance wider than the curve's own range is unfalsifiable; (7) the mean of a bimodal population ($D > 0.9$ vs. $D < 0.01$) is the share of broken sets, not a state; (8) a threshold must be *attainable* — an exact permutation test over three points has a floor of $1/3! = 0.167$, so a requirement of $p \le 0.05$ cannot be met by any behaviour of the model, and a perfectly monotone result ($\rho = +1.000$) was once recorded as failed; (9) a comparison point taken as the nearest element of a grid that does not bracket the target answers a question about the grid's edge, and a nearest element exists for every grid, so the defect is silent; (10) the range of n noisy values is itself random, so any threshold on spread must be compared with its own expected range before it is written down.

4. Results

4.1. The ledger

128 pre-registered parts across 26 rules (F2–F29): 66 passed, 48 failed, 13 unfalsifiable, 1 below its analytic bracket. Every rule is a pre-registered hypothesis split into parts; the table scores each rule

by the number of parts per verdict — P (passed), F (failed), U (unfalsifiable: the minimum detectable effect exceeded the written threshold), B (below the analytic bracket: the observation fell below the analytic fork, so the rule does not apply). Verdicts sit in an append-only ledger and are never revised.

Rule	Question	P/F/U/B
F2	onsets late in wakefulness; swing threshold	2/4/0/0
F3	damage delays sleep, lengthens episodes	8/0/1/0
F4	sleep deteriorates before the threshold (precursor)	3/3/1/0
F5	gradualness of the damaged fraction	0/1/0/1
F6	precursor: height, slope, slow noise	4/2/2/0
F7	duration within the two-day target	3/3/0/0
F8	serotonin burst as trigger	2/7/0/0
F10	episode composition: tail vs. time above window	3/2/0/0
F11	induction closes episodes	1/2/0/0
F12	deaf (refractory) period from defence	1/2/0/0
F13	independence of cell count J	2/2/0/0
F14	defence memory masked by noise	2/1/0/0
F15	refractory period exists, not from defence	3/0/0/0
F16	affine duration law, out-of-fit check	1/2/0/0
F18	who makes the precursor	2/2/2/0
F19	priming reaches the clinical target	4/0/0/0
F20	rhythm breaks via capture staircase	7/0/0/0
F21	third slow variable = population filling	2/1/0/0
F22	J via effective spread (substitution check)	1/1/0/0
F23	asymmetric defence follows $w(r)$	3/0/1/0
F24	bursty load beats smooth load	0/2/2/0
F25	three phases; band shape	3/2/0/0
F26	dopamine loop vs. static equivalent	2/2/0/0
F27	coexistence window of the load loop	2/3/0/0
F28	capacity loop, first edition (controls failed)	0/3/4/0
F29	capacity loop, matched differences	5/1/0/0

Not in the table: F9 and F17, whose objects were removed from the model as unnecessary. F9 tested slow-noise termination of episodes — the load held below the window boundary longer than the cell's repair time — and failed on all four parts; the reading was withdrawn, and the slow load noise was dropped from the model. F17 was the single-cell bench choosing between priming by amplification and by accelerating the first stage; the acceleration was inert (the first stage is equilibrated on episode timescales), so only amplification entered the model, where F19 tests it in full.

4.2. Established

1. *Duration is set by defence memory, not repair capacity.* 99.8 % of the within-set duration variance is explained by load history, 0.2 % by repair (exact decomposition). The law is affine in the defence memory τ_b (the slow stage of the two-stage chain):

$$\text{duration} = 1.28 + 0.97\,\tau_b \text{ (days)}, \tag{9}$$

   fitted below 100 h, checked at an out-of-fit 150 h; a shared-offset saturating form is rejected (F16b) though the quantitative out-of-fit prediction missed its 2 SE band (F16a). Passive repair misses the two-day target by 2–10× (the needed memory, ≈ 12 h, is several times shorter than every measured second-stage value, 24–88 h).

2. *Slow defence memory makes the pre-episode precursor:* removing it lowers the precursor in 107 of 174 sets, $p = 0.0015$; slow load noise is unnecessary.

3. *Three phases, not two.* A band of intact cells *more* vulnerable than their collapsed neighbours: 0.00732 ± 0.00042 of the population (17σ; ≈ 8.5 % on a fine grid); exactly zero with frozen defence; 1.82× wider under priming. Population-level hormesis, a model prediction, not a counting artefact.

4. *Priming (F19) reaches the clinical target:* the one-sided 95 % upper bound on the median episode duration is 1.51 d, inside the two-day target; the paired sign test gives $p = 4.0 \times 10^{-43}$ in the written direction and $p = 9.3 \times 10^{-13}$ for the reversed direction, and label permutation calibrates the criterion at 0.049; all four parts pass, replicated on 2000 sets. Unfitted: ceiling ×3 measured,

τ_p scanned; accelerating the first stage gives -1 to $-5\,\%$ vs. $-52\,\%$ for amplification, as predicted. Mechanism: priming raises the *top* of the window, not repair speed.

5. *Asymmetric defence (F23):* scanning $r = 0.33, 0.5, 1, 2, 3$ predicts $w = 0.441, 0.544, 0.705, 0.827, 0.878$ by (4); measured defence levels $0.0575, 0.0644, 0.0792, 0.0958, 0.1065$ — order exact ($\rho = +1$, permutation $p = 0.0083$). Damaged share falls $0.486 \to 0.457$ (Cochran–Armitage trend statistic 1.67 in the written direction); duration $7.74 \to 6.88$ d. Equal rise/decay constants were a *choice of regime*, not a simplification; r must be a model parameter unless measurement fixes it.
6. *Capacity loop (F29) converts frequent short episodes into rare long ones:* duration $13.1 \to 25.6$ d while the onset rate *falls* $0.042 \to 0.033$ per at-risk set-day; episodicity 0.75 of the open model; the broken state is held after release, monotone in release depth ($\rho = +1.000$, $p = 0.0083$); progression exceeds a capacity-matched static control by $+0.0231$ vs. MDE 0.0096, replicated as $+0.0223$, $+0.0211$ on two further seeds. The one failing part: the *shape* of risk in time is unchanged (risk trend 1.94 vs. threshold 2.23, F29c).
7. *Rhythm: a capture staircase* with rational rotation numbers and alternating cycle lengths on fractional steps (section 4.4).
8. *J enters only as effective spread of resilience;* the $J \to \infty$ limit exists with $1/J$ corrections (section 4.4).

4.3. Refuted

- *Grace-type load loop (F26, F27):* irreversibility latches at loop strength 1.11 while episodes extinguish at 0.18 — a sixfold structural gap, robust to a gentler protocol; no window with chronicity and episodicity together. The open model releases all broken sets ($0.9936 \to 0.0000$), the loop releases $0.24\,\%$. Reason: load is per-set, so the loop moves the whole population.
- *Capacity loop:* level and duration change, the shape of risk in time does not (above).
- *Bursty load does not beat smooth load (F24):* onset rate falls $40\,\%$ as correlation time grows at fixed mean and variance; the ever-damaged share saturates.
- *Serotonin burst (F8):* $26\,\%$ of onsets follow a burst, against an analytic expectation of $43\,\%$ and a pre-registered threshold of $80\,\%$ — a volume knob, not a trigger; no episode property changes.
- *Daily swing (F2b):* smooth $1.5\times$ growth, no gate; gradualness is from spread, not network (F5, observation *below* the analytic fork).
- *The episode does not remember its depth (F10a):* duration matches time-above-window (rank correlation 0.94 old / 0.89 current model); tail $\approx 6\,\%$ of the episode.
- *Band shape* $\varphi(\Phi^{-1}(1 - D))$ refuted twice with opposite signs (F25d,e) — width is not a function of D.
- *J-substitution check (F22b) failed:* the effective spread explains the measured drop but does not close the question (candidates: $1/J$ mean-dispersion, discreteness of D in the sleep feedback).
- Several precursor-slope parts were unfalsifiable at the chosen grid resolution (MDE larger than the written threshold).

4.4. Selected mechanisms

The staircase (F20). The deterministic switch under 24 h forcing, with damage inhibiting the sleep nucleus: the rotation number is pinned at rationals and jumps:

$$\begin{gathered} 1{:}1 \to 5{:}4 \to 3{:}2 \to 2{:}1 \to 3{:}1 \to 4{:}1 \to \cdots \to 9{:}1 \\ \to \text{no sleep}, \end{gathered} \tag{10}$$

transitions at $kD \approx 4.77$; 4.78; 4.79; 5.09; 5.16 mV; five steps (5:1–9:1) within 5.172–5.178 mV. A fine run: 52 steps in 0.9 mV, 34 with alternating cycle lengths (patterns $25, 24, 24, 47$ h at 4.772; $47, 25$ at 4.779; $46, 26, 24$ at 4.783); widths 0.3 mV (2:1), 8×10^{-3} (3:2), 5×10^{-6} (11:8). Capture staircase (Arnold tongues), *not* a period-doubling cascade: the period adds one skipped night; the discriminator is a capture on a non-power-of-two period (found). Limitation is calibration: $kD \le k$ identically, so at $k = 2$ mV the first step (4.77 mV) is unreachable at any damage; at $k = 5$ mV the model reaches it ($kD = 4.93$, $D = 0.999$).

Cell count J (F13, F22). Three paths: (1) the tail of the vulnerability distribution — closed by the fixed-edge grid (at $J = 32$ identical to the grid of earlier versions to the last digit); (2) quantisation of the episode threshold (0.625 at $J = 8$ vs. 0.516 at $J = 64$) — bypassed via the continuous variable; (3) the realised spread (section 2.5): 11 % in effective σ_v between $J = 16$ and 64, at log-sensitivity ≈ 2.5 $\Rightarrow \approx 25\,\%$ in measured quantities — exactly the range F13 found at 2000 sets. A discretisation can preserve mean and edges exactly and still miss the variance ($O(1/J)$): comparing runs at different J compares different models.

The capacity loop (F28, F29). The first edition collapsed on its own controls: "no increment with the loop open" is false — a bistable switch never releases a broken set, so the broken share rises on its own; the control passed at small n and failed as data accumulated. F29 therefore contains *no test against zero*: all claims are matched differences at equal damage level (section 5).

The band (F25). On a 600-node grid: frozen defence $\Rightarrow$ exact monotonicity (zero violations); live defence breaks order on 11.9 % of steps, worst violation 0.973 (full reversal); priming amplifies, does not cause ($A = 0$: 0.0184). Consequence: the closed-form damaged fraction $1 - F_v(v^*)$ — F_v the cumulative distribution of the load multiplier, v^* its threshold value — is illegal under live defence; any reduction needs an explicit three-configuration boundary layer.

5. Methodological lessons

In a model with absorbing states and a heterogeneous population, *no absolute null exists for any accumulation measure*: the damaged fraction rises on its own (bistable sets do not release) and the risk falls on its own (the vulnerable are absorbed first). Two of our controls asserted these baselines were zero; both were false and both *passed at small sample and failed as data accumulated* — a false null is passed by lack of power, since the threshold shrinks as $1/\sqrt{n}$ while the baseline is fixed. Every claim must be a matched difference between arms at the same damage level. The ten traps (section 3.3) are this lesson in ten costumes; the tenth cost us directly: $\mathbb{E}[\text{range}] = 1.693\,r$ for $n = 3$, so a "spread below 10 %" threshold was meaningless at $r = 12\,\%$ and became testable only at 2000 sets — where it failed and exposed the third path of J.

6. Scope and open questions

The model has no absorption-free stationary state: it is valid in windows short compared with the accumulation time — a model of the *generation and termination of an episode*, not of the course of illness. Everything measured here is moreover conditional on the bistability premise: a parameter set enters the study only if its cell already sits inside the window, and whether real PV cells do is *not* established by this work. The measured cell lines lie outside it (section 2.1) and entering requires repair weakened 1.6–22-fold; that weakening is the hypothesis under test, not a result of the tests. The capacity loop is the only mechanism tested that makes progression appear, and it does so by lengthening episodes, not by multiplying them. Open: (i) progression on illness scale needs a component that does not fully return — deliberately not added: a mechanism introduced to produce its target result is not a test; the falsifying rule comes first; (ii) regions coupled through narrow bottlenecks vs. the per-set homogeneity that killed the load loop; (iii) rhythm breakdown unreachable at $k = 2\,\text{mV}$ vs. first step 4.77 mV — miscalibration or another path; (iv) the band width is set by something third (likely the time near the tipping point) — testable against the mid-scale residence time, must agree on both scans or it is withdrawn; (v) the third slow variable behind the precursor (population filling has no own τ); (vi) the free term of (9) (1.3 d at zero memory) is unexplained by x; (vii) peroxiredoxin kinetics are from cell lines, not PV cells; (viii) dopamine and symptoms are outside the model by design.

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